Luis Fernando Pareja-Roman, Ph.D.

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Education

2019 Ph.D. Physical Oceanography | Rutgers University
2013 M.S. Meteorology | Universidad Nacional de Colombia
2010 B.S. Environmental Engineering | Universidad de los Andes, Colombia

Research Interests

Coastal Ocean Dynamics, Air-Sea Interaction, Coupled Atmosphere-Wave-Ocean Modeling

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Positions

2024 -	Assistant Research Professor, Marine and Coastal Sciences, Rutgers University
2022-2024	Research Associate, Marine and Coastal Sciences, Rutgers University
2021-2022	Postdoctoral Associate, Marine and Coastal Sciences, Rutgers University
2019-2021	Postdoctoral Researcher, Stevens Institute of Technology
2014-2019	Graduate Assistant, Marine and Coastal Sciences, Rutgers University
2012-2013	Graduate Fellow, Universidad Nacional de Colombia, Bogota, Colombia
2010	Environmental Engineer, Hidrocaribe Ltd, Cartagena, Colombia.

Peer-Reviewed Publications (* selected publications)

Pareja-Roman, L. F., Miles, T., & Glenn, S. (2024). Coastal Upwelling Modulates Winds and Air-Sea Fluxes, Impacting Offshore Wind Energy. *Frontiers in Energy Research*, 12. https://doi.org/10.3389/fenrg.2024.1470712

Pareja-Roman, L. F., Chant, R., Mazzini, P. L. F., & Cole, K. (2024). Circulation and retention of river plumes around capes. *Journal of Geophysical Research: Oceans*, 129, e2023JC020645. https://doi.org/10.1029/2023JC020645

***Pareja-Roman, L. F.**, Orton, P. M., & Talke, S. A. (2023). Effect of Estuary Urbanization on Tidal Dynamics and High Tide Flooding in a Coastal Lagoon. *Journal of Geophysical Research: Oceans*, 128(1), e2022JC018777. https://doi.org/10.1029/2022JC018777
AGU Highlight

Pareja-Roman, L. F., & Chant, R. (2023). Effects of a Changing Climate on the Physics of Estuaries. In M. Kennish, H. Paerl, & J. Crosswell (Eds.), *Climate Change and Estuaries* (pp. 696). CRC Press, Taylor & Francis. Book Chapter

Li, L., Wang, C., **Pareja-Roman, L. F.**, Zhu, J., Chant, R. J., & Wang, G. (2022). Effects of Typhoon on Saltwater Intrusion in a High Discharge Estuary. *Journal of Geophysical Research: Oceans*, 127(8), e2021JC018206. https://doi.org/10.1029/2021JC018206

Li, L., Zhu, J., & **Pareja-Roman, L. F.** (2021). Calculating salinity variance fluxes using isohaline coordinates. *Estuarine, Coastal and Shelf Science*, 254, 107311. https://doi.org/10.1016/j.ecss.2021.107311

***Pareja-Roman, L. F.**, Chant, R. J., & Sommerfield, C. K. (2020). Impact of Historical Channel Deepening on Tidal Hydraulics in the Delaware Estuary. *Journal of Geophysical Research: Oceans*, 125(12), e2020JC016256. https://doi.org/10.1029/2020JC016256

Li, L., Zhu, J., Chant, R. J., Wang, C., & **Pareja-Roman, L. F.** (2020). Effect of Dikes on Saltwater Intrusion Under Various Wind Conditions in the Changjiang Estuary. *Journal of Geophysical Research: Oceans*, 125(7), e2019JC015685. https://doi.org/10.1029/2019JC015685

***Pareja-Roman, L. F.**, Chant, R. J., & Ralston, D. K. (2019). Effects of Locally Generated Wind Waves on the Momentum Budget and Subtidal Exchange in a Coastal Plain Estuary. *Journal of Geophysical Research: Oceans*, 124(2), 1005-1028. https://doi.org/10.1029/2018JC014585

Pareja-Roman, L. F., Diaz-Guevara, D. C., Rodriguez-Tobar, A. T., Villegas, N. L., & Perez-Santos, I. (2013). Análisis del transporte y bombeo de Ekman en el Caribe colombiano entre 1999 y 2009. *Boletín Científico CIOH*, 31, 3-12. https://doi.org/10.26640/22159045.248

Conference Presentations

Miller, M., Wilkin, J., Zheng, Q., **Pareja-Roman, L.F.**, Braddock, S. (2024). Modeling the Impact of Wind Farms on Surface Currents and Marine Atmospheric Boundary Layer Turbulence Structure. NAWEA / WindTech 2024 Conference [Talk]

Pareja-Roman, L.F., Miles, T., Glenn, S (2024). How Does Coastal Upwelling Influence Offshore Wind Energy? NAWEA / WindTech 2024 Conference [Poster]

Kim, J. Miles, T., Crowley, M., **Pareja-Roman, L.F.**, Garzio, L., Nazzaro, L., Glenn, S. The Rutgers University Weather Research and Forecasting Model (RU-WRF). NAWEA / WindTech 2024 Conference [Poster]

Cassin, R., Miles, T. N., & **Pareja-Roman, L. F**. (2024). Investigating Potential Impacts of Offshore Wind Turbine Foundations on the Mid-Atlantic Bight Cold Pool. 2024 Ocean Sciences Meeting [Poster]

Mazzini, P. F., **Pareja-Roman, L. F.**, Cole, K. L., Chant, R. J., & Hunter, E. J. (2024). High-resolution observations of a river plume interacting with a cape: impacts on hydrography and along-shelf transport. 2024 Ocean Sciences Meeting. [Poster]

Pareja-Roman, L. F., Chant, RJ., Mazzini, P.L., Colle, K. (2024). The Role of Capes on River Plume Mixing, Retention, and Transport in the Coastal Ocean: Idealized Modeling. AGU Ocenan Sciences Meeting. New Orleans, USA. [Talk]

Pareja-Roman, L. F., Chant, RJ., Mazzini, P.L., Colle, K. (2023). Circulation, Transport, and Retention of River Plumes Around Capes. II Latin American Physics of Estuaries and Coastal Oceans (LAPECO). Panama City, Panama. [Poster]

Pareja-Roman, L. F., and Wilkin, J. (2023). Advancing a Coupled Ocean-Wave Modeling System to Predict Coastal Storm Impacts. Summer school with WAVEWATCHIII ®. Institut Français de Recherche pour l'Exploitation de la Mer (IFREMER). Brest, France. [Poster]

Pareja-Roman, L. F., Miles, T., Glenn, S. (2023). Effect of Coastal Upwelling on Air-Sea Interaction and Offshore Wind in New Jersey. Rutgers Offshore Wind Symposium. [Talk]

Pareja-Roman, L. F., Orton, PM.; Talke, SA. (2021). Impact of Urbanization on Tide and Flood Dynamics in Jamaica Bay, New York: Realistic and Analytical Modeling. American Geophysical Union (AGU) Fall Meeting, USA. [Talk]

Pareja-Roman, L. F., Chant, RJ. (2018). Shifts in tidal and salt dynamics in Delaware Bay. Physics of Estuaries and Coastal Seas Conference (PECS). Galveston, Texas. USA. [Talk]

Pareja-Roman L. F., and Chant, R. (2017). Tidal variability of the wave energy budget in Delaware Bay. 24th Biennial Conference of the Coastal and Estuarine Research Federation (CERF). Providence, RI. USA. [Poster]

Pareja-Roman L. F., and Chant, RJ (2017) Modeling the wind and wave-driven circulation in Delaware Bay. Gordon Research Conference, Coastal Ocean Dynamics. Biddeford, ME. USA. [Poster]

Pareja-Roman, L. F. (2016) Energy Fluxes and Water Level in a Tidal Estuary. International Symposium of Environmental Hydraulics. International Association for Hydro-Environment Engineering Research (IAHR). University of Notre Dame, IN, USA. [Talk]

Pareja-Roman L. F., and Chant, RJ. (2016). Delaware Bay- hydrodynamics in the presence of wind waves: insight from a coupled wave-current numerical model. Physics of Estuaries and Coastal Seas Conference (PECS). Scheveningen, Netherlands. [Poster]

Chant, RJ., **Pareja-Roman L.F**., Geyer, WR., Ralston, DK, Sommerfield, CK, and Quirk, T. (2015). The wave climate in a highly engineered estuarine basin. 23rd Biennial Conference of the Coastal and Estuarine Research Federation. Portland, OR. [Poster]

Pareja-Roman L. F. (2013). Spatiotemporal variability of air-sea heat and momentum fluxes in the Colombian Caribbean basin. CICESE Wave Research Group Seminar. Department of Physical Oceanography. Ensenada, Mexico. [Talk]

Díaz, D.; **Pareja-Roman L. F.**, Rodríguez, A., Villegas, N. (2013). Sensible and Latent Flux variability in the Colombian Pacific Basin – Panama Bight Region. XV Latin-American Congress of Marine Sciences (COLACMAR). Punta del Este, Uruguay. [Poster]

Pareja-Roman L. F., Villegas, N; Díaz, D; Rodríguez, A. (2013). Analysis of Ekman Transport and Pumping in the Colombian Basin, 1999 to 2009. XVI National Seminar of Marine Science and Technology. Cartagena, Colombia. [Poster]

Pareja-Roman L.F. (2012). Momentum flux variability in the Colombian Basin, Caribbean Sea. Marine Biology Seminar. Universidad Jorge Tadeo Lozano. Bogota, Colombia. [Talk]

Seminars

Pareja-Roman L. F. (2024). Effect of ROMS-WRF Atmosphere-Ocean Coupling on Forecasting Skill: Preliminary Results. Polar and Coastal Ocean Dynamics Seminar. Rutgers University.

Pareja-Roman L. F. (2024). The Anatomy of Air-Sea Turbulent Fluxes. Polar and Coastal Ocean Dynamics Seminar. Rutgers University.

Pareja-Roman L. F. (2024). Modeling Air-Sea Fluxes and Wind Effects on Currents under Hurricane Irene (ROMS-WRF) – Preliminary Results. 7320- Ocean Dynamics and Prediction Branch Seminar. U.S. Naval Research Laboratory, Stennis, MS.

Pareja-Roman L. F. (2024). Impact of Cool Skin Temperature, Relative Wind Speed, and Surface Roughness on Coastal, Hurricane-driven Circulation. Polar and Coastal Ocean Dynamics Seminar. Rutgers University.

Pareja-Roman L. F. (2022). Estimates of Ocean Mixing by Offshore Wind Farms in the Mid Atlantic Bight. Polar and Coastal Ocean Dynamics Seminar. Rutgers University.

Pareja-Roman L. F. (2021). Modeling Natural and Anthropogenic Processes in the Coastal Ocean. Jose Celestino Mutis Lecture Series. Universidad Nacional de Colombia. Bogota, Colombia.

Pareja-Roman L. F. (2019). Dredging the Delaware Estuary from 1848 to now: Effects on Salinity and Tides. Civil and Environmental Engineering Seminar Series. Rutgers University.

Pareja-Roman L. F. (2018). Modeling Waves and Tides in Delaware Bay. Woods Hole Group – A CLS Company. Bourne, MA.

Pareja-Roman L. F. (2018). Evaluation of a Vortex Force Wave Model in an Urban Estuary. Civil, Environ., and Ocean Engineering Seminar. Stevens Institute of Technology.

Pareja-Roman L. F. (2015). Environmental Fluid Mechanics and its Role in Climate Change Adaptation (2015). 65th Anniversary of the Society of Architects and Engineers. Cartagena, Colombia.

Research Projects (* as a PI/co-PI)

*Vineyard Wind (2024-2025). Assessing the hydrodynamic impacts of ocean and wind wakes associated with offshore wind turbine structures in the Nantucket Shoals region. Rutgers University.

*NJ-BPU (2024-2025). Model intercomparisons of RUWRF to NOW-23 and evaluation of model impacts on grid efficiency and safety. Rutgers University.

*NJ-DEP (2023-2024). Impacts of wind turbines and foundations on the Mid-Atlantic Cold Pool. Rutgers University.
U.S. Coast Guard (2024-2025). Modeling the impact of wind farms on surface currents and marine boundary layer conditions. Rutgers University.
NSF-AGS (2023-2025). Process mechanics of cloudiness transitions in subtropical marine boundary layers. Rutgers University.
USGS (2023-2025). Advancing a coupled ocean-wave modeling system to predict coastal storm impacts. Rutgers University.
NJ-BPU (2022-2023). Targeted stakeholder engagement & product development using the RU-WRF model database. Rutgers University.
NJ-BPU (2021-2022). Operations & maintenance of the RU-WRF model and expanded research in support of offshore wind. Rutgers University.
NSF-OCE (2021-2023). Plume separation from the coastal wall, vorticity generation, and freshwater retention. Rutgers University.
NSF-PREEVENTS (2019-2021). Geomorphic and climatic drivers of coastal flood risk. Stevens Institute of Technology.
NSF Coastal SEES (2014-2019). Towards estuarine sustainability in the Anthropocene. Rutgers University.

Teaching

Certification:
- Rutgers Academy for the Scholarship of Teaching and Learning (2018)
- Sample teaching material: https://github.com/lfpareja/lfpareja.github.io/tree/master/teaching

Instructor:
- Numerical Modeling of the Ocean and Atmosphere I, Rutgers University (Fall 2024)
- Climatology, Universidad Nacional de Colombia (Fall 2013)

Invited Lectures:
- Ocean Methods and Data Analysis, Rutgers University (Spring 2023, 2024)
- Sea Level Science, Stevens Institute of Technology (Spring 2020)
- Operational Ocean Modeling and Visualization, Rutgers University (Fall 2022)
- Hydrogeology, Universidad Nacional de Colombia (Fall 2013).

Teaching Assistant:
- The Water Planet, Rutgers University (Spring 2019, 2020)
- Coastal Ocean Dynamics, Rutgers University (Fall 2017)
- Introduction to Oceanography, Rutgers University (Spring 2017)
- Physical Oceanography, Rutgers University (Fall 2015)
- External Geodynamics, Universidad Nacional de Colombia (Spring, Fall 2012)
- Introduction to Civil Engineering, Universidad de los Andes (Spring 2010)
- Introduction to Environmental Engineering, Universidad de los Andes (Spring 2010)

Mentoring

Postdoctoral Associate (co-advised):
- Qiuxuan Zheng, Meteorology, Rutgers University

Ph.D. Students (co-advised):

- Stephanie Braddock, Meteorology, Rutgers University
- Leah Hopson, Oceanography, Rutgers University
- Feng Ye, Industrial and Systems Engineering, Rutgers University

Undergraduate Scholars:

- Julia O'Neil, B.S. Meteorology, George H. Cook Scholar
- Angela Poulos, B.S. Meteorology, Student of the Year, Amelia Ruggles and Philip Alampi Scholar.

NSF Research Experience for Undergraduates:

- Ryan Cassin, University of Rhode Island (at Rutgers)
- Nasir Perera-Olivo, Brown University (at Rutgers)

Workshops and Internships

2023 Summer School with WAVEWATCH III. IFREMER, Brest, France.
2021 Meteorology in a Round-The-World Regatta. Universitat de Barcelona, Spain.
2016 COAWST (ROMS, SWAN, WRF, CSTMS) Workshop, USGS Woods Hole.
2013 Geophysical Fluid Dynamics Internship, CICESE, Ensenada, Mexico
2010 Summer School. Russian State Hydrometeorological University, Saint Petersburg, Russia

Oceanographic Field Experience

Drakes Bay, California (R/V Rachel Carson). NSF-OCE. Shipboard surveys of velocity and salinity around capes, 1 week (2022)

New York Harbor and Newark Bay, New York/New Jersey (DCV Howard | US Army Corps of Engineers). Inspection of coastal structures and discussion of regional research needs, 1 day (2019)

Raritan Bay (R/V Rutgers). NSF Coastal SEES. Drifter deployment for survey of surface ocean mixing, 1 day (2016).

Newark Bay (R/V Sharp). NSF Coastal SEES. Tidal survey and shipboard measurements of estuarine frontogenesis, 1 day (2016).

Jamaica Bay, New York (R/V Arabella). Jamaica Bay Observing System: Groundwork for LTER. Assisted with mooring recovery (ADVs and Oxygen/Nutrient sensors). 1 day (2015).

Delaware Bay, New Jersey/Delaware (R/V N.A.) NSF Coastal SEES. Sidescan benthic survey and mooring recovery. 1 day (2015).

Chesapeake Bay Plume, Virginia (R/V Arabella). Circulation and Mixing in a Coastally Trapped River Plume. Deployment of moorings, profilers, and Autonomous Underwater Vehicles. 4 days (2015)

Delaware Bay (R/V Sharp). NSF Coastal SEES. Mooring deployment and recovery, tidal surveys. 4 days (2014)

Islas del Rosario, Colombia (R/V N.A). Oceanographic surveys of seagrass meadows and coral reefs; reconnaissance field work. 5 days (2012)

Neva Estuary, Gulf of Finland, Russia (R/V Centaurus II). Marine weather survey. Russian State Hydrometeorological University. 2 days (2010)

Awards and Honors

2023	Highlight by the American Geophysical Union, JGR-Oceans
2019	Graduate Development Award, Rutgers University
2017	O'Brien Fund Travel Award, Rutgers University
2012	Outstanding Graduate Student Fellowship, Meteorology (2 years)
2010	Top Scores in Environmental Engineering Examination, Saber PRO, Colombia
2006	Full Scholarship, Environmental Engineering (4 years)

Service and Outreach

2016-2019	Founder, Coastal Ocean and Estuarine Processes Seminar at Rutgers University (now Polar and Coastal Ocean Dynamics Seminar)
2016-2017	Secretary, Oceanography Graduate Student Association (OGSA)
2015	Runner, NJ Shore Bowl (competition for high school students; grades 9-12)

Press and media coverage

Phys.org, AGU News, Baruch College Susan Locke Capstone Blog, and others.

Peer Reviewing

Journal of Geophysical Research: Oceans, Journal of Coastal Research, Continental Shelf Research.

Professional Affiliations

American Geophysical Union (AGU)
American Meteorological Society (AMS)
U.S. National Maritime Historical Society (NMHS)
Professional Environmental Engineer (Colombia)